

$$\Phi_E = EA \cos \theta$$

(ii). General Formula (Non-uniform Field or curved surface)

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

(iii) Gauss's law (for closed surface).

$$\Phi_E = \frac{\Phi_{\text{enclosed}}}{\epsilon_0}$$

ϵ_0 = permittivity of free space
 $= 8.854 \times 10^{-12} \text{ F/m}$

Electric potential

$$V = \frac{U}{q}$$

$\rightarrow$ electric potential energy
 $\rightarrow$ Charge.

$\hookrightarrow$ Electric potential

or Electric potential

also Equal to

$$V = k \frac{q}{r}$$

Electric potential : an amount of work needed to move a unit positive charge from reference point to the point of Electric Field.

Potential Energy

$$V = \frac{U}{q}$$

$$U = VQ$$

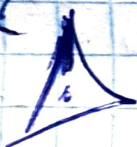
↳ potential energy

definition

↳ is directly proportional to the product of potential and charges.

$$U = \frac{kq_1q_2}{r}$$

$$U = k_e \left(\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \dots \right)$$



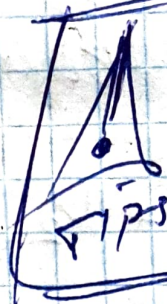
Electric Dipole: system consisting of two equal and opposite point charges (+Q and -Q) separated by very small distance.

system charge is zero.

S.I unit (Cm).

Electric potential due to continuous charge distribute

$$V = k \int \frac{dq}{r}$$



we consider the sign of charge

electric potential at the surface of the spheres
equal to each other.

$$V_1 = V_2$$

$$k_e \frac{q_1}{r_1} = k_e \frac{q_2}{r_2}$$

$$\boxed{\frac{q_1}{r_1} = \frac{q_2}{r_2}}$$

Capacitance and Dielectrics.

Definition: Capacitor is directly proportional to the charge and inversely proportional to the electric potential magnitude.

$$C = \frac{Q}{\Delta V} \quad \text{S.I. Farad (F)}$$

Capacitance: defined as ratio of the magnitude of charge on either conductor to the magnitude of the potential difference between conductors.

$$\Delta V = k_e \frac{Q}{r}$$

$$k_e = \frac{1}{4\pi\epsilon_0}$$

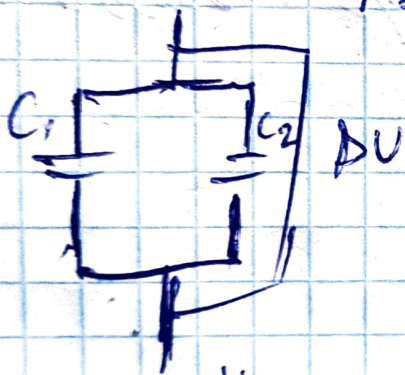
$$C = \frac{Q}{k_e \frac{Q}{r}}$$

$$\Rightarrow C = \frac{Q}{\frac{1}{4\pi\epsilon_0} \frac{Q}{r}}$$

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{4\pi\epsilon_0 r}} = \frac{4\pi\epsilon_0 r}{1}$$

$$C = 4\pi\epsilon_0 r$$

Where ϵ_0 : permittivity $= 8.854 \times 10^{-12} \text{ C}^2/\text{Vm}$
 r : Distance b/w conductor & Charge
parallel-plate capacitor



$$C = \frac{Q}{\Delta V}, \quad C_1 = \frac{Q_1}{\Delta V}, \quad C_2 = \frac{Q_2}{\Delta V}$$

$$\Delta V : \text{unchangeable} \quad \Delta V_1 = \Delta V_2 = \Delta V$$

$$Q_T = Q_1 + Q_2$$

$$C_T = C_1 + C_2$$

$$C_T = \frac{Q_1}{\Delta V} + \frac{Q_2}{\Delta V} \quad \Rightarrow \quad C_T = C_1 + C_2 + \dots$$

$$Q_T = C_T V \Rightarrow Q_T = C_T V$$

$$Q = Q_1 + Q_2$$

$$C_T X_T = X(C_1 + C_2)$$

$$\boxed{C_T = C_1 + C_2} \quad \text{Capacitors in parallel.}$$

also it can be expressed in terms of Area.

$$E = \frac{P}{\epsilon_0}$$

ϵ_0 : permittivity

$P = Q/A$ Surface density Area

$$E = \frac{Qd}{\epsilon_0 A}$$

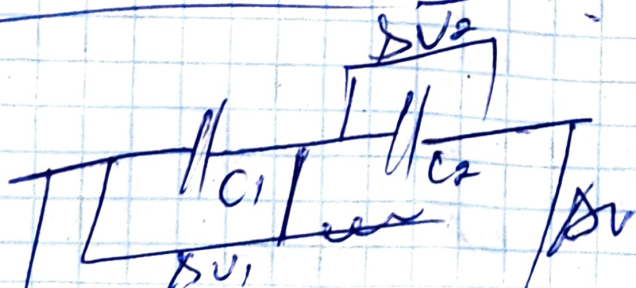
$$\Delta V = Ed$$

$$\Delta V = \frac{Qd}{\epsilon_0 A}$$

$$C = \frac{Q}{\Delta V} \Rightarrow C = \frac{Q}{\frac{Qd}{\epsilon_0 A}}$$

$$\boxed{C = \frac{\epsilon_0 A}{d}}$$

Series Combination



$$Q_1 = Q_2 = Q_T$$

$$Q = C \Delta V \Rightarrow Q_T = C_T \Delta V_T$$

$$C_T \Delta V_T = \Delta V_T = V_1 + V_2 \dots$$

$$V_T = \frac{Q}{C_T}$$

$$\frac{Q}{C_T} = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{Q}{C_T} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

$$\boxed{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}} \text{ Series combination.}$$

~~etc~~

Current and Resistance

Current

The current is the rate at which the charge flows through the surface.

$$I = \frac{Q}{t}$$

S.I is Ampere (A)

The current is conventional to assign to the current the same direction as the flow of positive charges.

this is drift speed: the speed of charge carriers.

$$I = \frac{\Delta Q}{\Delta t} = nqV_d A.$$

Ex: $A = 3.31 \times 10^{-6} \text{ m}^2$ $I = 10.0 \text{ A}$ $ng = 12.$

Drift Speed?

$$I = nqV_d A$$

$$V_d = \frac{I}{nqA}$$

$$V_d = \frac{10.0 \text{ A}}{12 \times 3.31 \times 10^{-6} \text{ m}^2}$$

Resistor

Electric Resistance is the measure of opposition to the flow of electric current.

$$R = \frac{\rho l}{A}$$

$$S.I = \text{ohm } (\Omega).$$

where

ρ = resistivity of material.

$$\rho = \frac{1}{\sigma} \quad ; \quad \sigma : \text{Conductivity of material}$$

Electromotive Force

The ~~emf~~ $\mathcal{E}$ of Battery voltage is the maximum possible voltage can provide Btwn its terminals

$$\boxed{\mathcal{E} = IR + Ir} \rightarrow \text{emf } \mathcal{E}$$

$$\mathcal{E} = I(R+r)$$

$$\boxed{I = \frac{\mathcal{E}}{R+r}}$$

where R : Internal Resistance

r : Load Resistance

$$\boxed{\mathcal{E} = \frac{W}{Q}} \rightarrow \text{General formula}$$

where: $\mathcal{E}$: Electromotive force

W = work done

Q = Charge.

S.I. units (J/C)

Ohm's Law

State that: Current through a conductor between two points is directly proportional to the voltage across those points.

Temperature remain constant

$$\begin{aligned} V &= I \times R \\ I &= \frac{V}{R} \\ R &= \frac{V}{I} \end{aligned}$$

$$\begin{aligned} P &= VI \\ &= (IR)I \end{aligned}$$

$$\boxed{P = I^2 R}$$

~~Potential Difference~~ :

Energy Stored

$$U_E = \frac{1}{2} CV^2.$$

Electric Dipole : This is vector Quantity defined as the product of the charge (q) and the displacement vector (d) pointing from the negative charge to the positive charge.

$$P = qd.$$

d : is vector separating the charges.